Chapter 2: Continuous Time Markov Chain

October 6, 2025

Introduction

A Continuous-Time Markov Chain (CTMC) is a stochastic model that generalizes discrete-time Markov chains (DTMC) by allowing transitions between states to occur at continuous moments in time.

Introduction

Here are some key concepts related to CTMCs:

State: A CTMC consists of a set of states, similar to discrete-time Markov chains.

Markov Process: The fundamental property is the Markov property.

Holding Time: The time spent in a state before making a transition is a random variable, typically modeled by an exponential distribution.

Transition Rate (or Generator): CTMC allows transitions at any continuous moment, with a transition rate matrix Q. The element q_{ij} in this matrix represents the rate of moving from state i to state j.

Exponential Distribution

Definition

A random variable X is distributed exponentially $X \rightsquigarrow \mathcal{E}(\lambda)$ with rate $\lambda > 0$, if X has the probability density function:

$$f(t) = \begin{cases} \lambda e^{-\lambda t}, t \geq 0 \\ 0, t < 0 \end{cases}$$

and the cummulative distribution function,
$F(t) = \left(1 - e^{-\lambda t}\right) \mathbb{I}_{t \geq 0}$. So $\overline{F}(t) = e^{-\lambda t}, t \geq 0$.

Exponential Distribution

In addition,

$$\begin{aligned}\mathbb{E}[X] &= \frac{1}{\lambda}, \mathbb{E}\left[X^2\right] = \frac{2}{\lambda^2}, Var(X) = \frac{1}{\lambda^2} \\ \mathbb{E}\left[e^{-tX}\right] &= \frac{\lambda}{\lambda - t} \text{ for } t < \lambda.\end{aligned}$$

Definition

A random variable X is said to be memoryless if

$$\mathbb{P}(X > s + t / X > s) = \mathbb{P}(X > t) \forall s, t \geq 0.$$

Proposition (Memoryless Property of the exponential)

The exponential variable has the memeoryless property.

Exponential Distribution

Property

Proposition: The minimum of two independant exponential r.v follows an exponential law: let $X_1 \sim \mathcal{E}\left(\lambda_1\right)$ and $X_2 \sim \mathcal{E}\left(\lambda_2\right)$ two independent varaibles, so

$$X = \min\left(X_1, X_2\right) \sim \mathcal{E}\left(\lambda_1 + \lambda_2\right).$$

In addition,

$$\mathbb{P}\left(\min\left(X_1, X_2\right) = X_1\right) = \mathbb{P}\left(X_1 \leq X_2\right) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Exponential Distribution

Property

The sum of n independent exponential r.v with the same rate follows the Gamma law: let $X_1, X_2, ..., X_n$ i.i.d of $\mathcal{E}(\lambda)$, then:

$$\sum X_i \sim Gamma(n, \lambda)$$

where the dpf of $Gamma(n, \lambda)$ is

$$\begin{aligned} f(x; n, \lambda) &= \frac{\lambda^n}{\Gamma(n)} x^{n-1} e^{-\lambda x} \mathbb{I}_{x \geq 0} \\ &= \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x} \mathbb{I}_{x \geq 0}. \end{aligned}$$

Poisson Process

A Poisson process is a continuous-time Markov chain $\{X_t, t \geq 0\}$, which counts the number of arrivals of an event that occur randomly over time at a constant rate $\lambda > 0$.
A flow of random events can be described mathematically in two different ways:

1. The number of events X_t occurring in $[0, t]$ and seek to determine the probability law of this discrete random variable. The process $\{X_t, t \geq 0\}$, is called the "*counting process*".
2. The times intervals that separate the instants of occurrence of two consecutive events is called "*inter-arrival time*". These are independent and identically distributed r.v with law $\mathcal{E}(\lambda)$. This process is called the "*birth process*".

Poisson Process

Definition

A Poisson process having rate λ is a sequence of events such that

$$
\begin{aligned}
\mathbb{P}\left(X\left(t\right)=k\right) &= p_k\left(t\right)=\frac{\left(\lambda t\right)^k}{k!}e^{-\lambda t}, \lambda>0, k\geq 0;\\
E\left[X\left(t\right)\right] &= \lambda t \text{ and } Var\left[X\left(t\right)\right]=\lambda t;\\
E\left[e^{tX}\right] &= e^{\lambda\left(e^t-1\right)}.
\end{aligned}
$$

Poisson Process

Proposition: The sum of n independent Poisson r.v follows the Poisson law: let $X_1, X_2, ..., X_n$ i.i.d of $\mathcal{P}(\lambda_i)$, so:

$$\sum_{i=1}^{n} X_i \sim \mathcal{P}\left(\sum_{i=1}^{n} \lambda_i\right).$$

Introduction

The processes in question can generally be used to write the temporal evolution of the size of a population of a given type. They are widely used to model waiting phenomena or systems subject to repairable failures. They are obtained by superimposing a birth process and a death process. This Markov process X_t represents the size of a population at time t. These are stochastic processes with continuous time and discrete states space $S = \{0, 1, 2, ...\}$. They are characterized by two important conditions:

- Memoryless;
- transitions are only possible to one or other of the neighboring states, from state n the possible transitions are $n - 1$ and $n + 1$ with $n \geq 1$.

Birth Process

The birth process characterizes the arrival (or appearance) of an individual within the population, according to a certain law. Let $\{N(t), t \geq 0\}$, where $N(t)$ is the number of individuals in the population at date t.

Between t and $t + \Delta t$, knowing that i individuals already exist in the population, the probability of appearance of:

a) an individual (one birth), is $p_{i,i+1}(\Delta t) = \lambda_i \, \Delta t + o\Delta t,$

b) no individual (no birth), is $p_{i,i}(\Delta t) = 1 - \lambda_i \Delta t + o \, \Delta t,$

c) two or more individuals (more than one birth), is negligible $p_{i,i+k}(\Delta t) = o\Delta t, k \geqslant 2.$

Here, λ_i represents the **infinitesimal transitions rates** (generators) of **birth** (or appearance or the rate of growth).

Death Process

The process of death characterizes the disappearance (also known as death or departure) of an individual.

Let $\{N(t), t \geq 0\}$, with states space $S = \{0, 1, 2, ...\}$.

Knowing that i individuals already exist in the population, between t and $t + \Delta t$, the probability of disappearance of:

a) of one individual (one death), is
$p_{i,i-1}(\Delta t) = \mu_i \Delta t + o\Delta t$,

b) no individual (no death), is
$p_{i,i}(\Delta t) = 1 - \mu_i \Delta t + o\Delta t$,

c) two or more individuals (more than one death), is negligible $p_{i,i-k}(\Delta t) = o\Delta t, k \geqslant 2$.

Where μ_i is the **infinitesimal transitions rates** (generators) **of disappearance** or the rate of decrease from the state i with $\mu_0 = 0$.

Birth and Death Process

Let $X_t = N(t)$ be the size of a population at time t (number of individuals present). We have that $p_{i,j}(t) = \mathbb{P}(N(t+s) = j / N(s) = i)$ does not depend on s (the process is homogeneous), so

$$\begin{aligned}
p_{i,i+1}(\Delta t) &= \lambda_i \Delta t + o\Delta t, \\
p_{i,i-1}(\Delta t) &= \mu_i \Delta t + o\Delta t, \\
p_{i,i}(\Delta t) &= 1 - (\lambda_i + \mu_i)\Delta t + o\Delta t, \\
p_{i,j}(\Delta t) &= o\Delta t \text{ if } |i - j| \geq 2 \text{ and } p_{i,j}(0) = \delta_{ij} = \begin{cases} 1, i = j \\ 0, i \neq j \end{cases}
\end{aligned}$$

with $\lambda_i > 0, \mu_i > 0$ and $\mu_0 = 0$.

Birth and Death Process

Transitory State

Let the transitions probabilities (or states probabilities) $p_n(t) = \mathbb{P}(N(t) = n), \forall n \geq 0$ during the interval time $[0, t]$. The transitory state is described by

$$\mathbb{P}(t + \Delta t) = \mathbb{P}(t) \times M$$

where M is transition matrix obtained from representative graph.

Birth and Death Process

Transitory State

Letting $\Delta t \to 0$, we obtain the following **Kolmogorov equations** system:

$$\begin{cases} p_0'(t) = -\lambda_0 p_0(t) + \mu_1 p_1(t) \\ p_1'(t) = \lambda_0 p_0(t) - (\lambda_1 + \mu_1) p_1(t) + \mu_2 p_2(t) \\ - - - - - - - - - - - - \\ p_n'(t) = \lambda_{n-1} p_{n-1}(t) - (\lambda_n + \mu_n) p_n(t) + \mu_{n+1} p_{n+1}(t), n \geq 1. \end{cases} \quad (1)$$

Remark: If $S = \{0, 1, ..., K\}$, then $\lambda_K = 0$. Hence, $p_K'(t) = \lambda_{K-1} p_{K-1}(t) - \mu_K p_K(t)$.

Birth and Death Process

Stationory state

In state of equilibrium, also known as steady state, the behavior of the process is independent of the time and of the initial state; i,e, $p_n = \lim\limits_{t\to\infty} p_n(t), n = 0, 1, 2...$which is the stationary distribution of the process under study. Therefore, $\lim\limits_{t\to\infty} p_n'(t) = 0.$ Then, these probabilities satisfy the following **balance equations** *or* **statistical equilibrium system** (obtained from (1) by taking):

$$\begin{cases} 0 = -\lambda_0 p_0 + \mu_1 p_1 \\ 0 = -(\lambda_1 + \mu_1) p_1 + \lambda_0 p_0 + \mu_2 p_2 \\ - - - - - - - - - - - - \\ 0 = -(\lambda_n + \mu_n) p_n + \lambda_{n-1} p_{n-1} + \mu_{n+1} p_{n+1}, n \geq 1. \end{cases} \tag{2}$$

To solve this system, we add $\sum\limits_{n=0}^{\infty} p_n = 1.$

Birth and Death Process

Stationory state

From (2), we obtain

$$\begin{aligned} p_1 &= \frac{\lambda_0}{\mu_1} p_0 \\ p_2 &= \frac{\lambda_0 \times \lambda_1}{\mu_1 \times \mu_2} p_0 \\ & - - - - - - \\ p_n &= \frac{\lambda_0 \times \lambda_1 \times \ldots \times \lambda_{n-1}}{\mu_1 \times \mu_2 \times \ldots \times \mu_n} p_0 \end{aligned} \quad (3)$$

To find p_0, we use the normalization equation.

Birth and Death Process

Stationory state

We get

$$p_0 = \left[1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0\lambda_1}{\mu_1\mu_2} + \ldots + \frac{\lambda_0 \times \lambda_1 \times \ldots \times \lambda_{n-1}}{\mu_1 \times \mu_2 \times \ldots \times \mu_n} + \ldots\right]^{-1} \quad (4)$$

Therefore, for a stationary distribution to exist, the sum between [ ] must converge. This is always the case if the state space of the process under study is finite.

Remark$_1$: When the sum in question is not convergent, $p_n = 0, \forall n \geq 0$.

Remark$_2$: We have that $p_0(0) = 1$ (population is empty) and $p_n(0) = 0$.